



JERUSALEM COLLEGE OF ENGINEERING, CHENNAI – 600 100
(An Autonomous Institution Affiliated to Anna University, Chennai)

Reg. No. :

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Question Paper Code : 235026

B.E./B.Tech. DEGREE END SEMESTER EXAMINATIONS, APRIL/MAY 2025

Sixth Semester

Computer Science and Engineering (Artificial Intelligence and Machine Learning)

JAL1603 – TIME SERIES ANALYSIS AND FORECASTING

(Common to Computer Science and Engineering (Cyber Security) and Computer Science and Business Systems)

(Regulation 2021)

Time : Three hours

Maximum : 100 marks

Answer ALL questions

PART A – ($10 \times 2 = 20$ marks)

1. What is meant by internal structures of time series?
2. Give the generic approach of a time series analysis.
3. Define stationary time series.
4. What is data transformation in time series analysis and forecasting?
5. Write short notes on least squares method.
6. State Cochrane–Orcutt method.
7. List the types of stationarity of a time series.
8. What are Yule-Walker equations?
9. What is an artificial neural network (ANN)?
10. Define periodogram?

PART B – (5 × 16 = 80 marks)

11. (a) What are the different types of data used in analytics. Explain their purpose with suitable examples.

Or

- (b) Explain about the role of data warehouses, data cleaning and imputation in time series analysis.

12. (a) What is the purpose of data transformations in time series data? Describe the different types of data transformations.

Or

- (b) Give the importance of monitoring the performance of a forecasting model.

13. (a) Write a note on statistical procedure for estimation of confidence intervals on individual regression coefficients and the mean response.

Or

- (b) Explain in detail about the various techniques available for variable selection in regression.

14. (a) Explain in detail with relevant representations about Finite order autoregressive processes.

Or

- (b) Illustrate ARIMA models with examples.

15. (a) Explain in detail about the multivariate time series models.

Or

- (b) Describe the structure of a typical neural network used in forecasting. How are features and time lags selected for multivariate time series?
